Exercises Day 1

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You already made exercises 1 and 2 in the slides. For exercises 3 to 7, you will do them in RStudio. Please make sure that you have R and RStudio installed on your laptop.

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Exercise 1: selection within a vector

a. Create a vector named `vec` that consists of the numbers from 11 to 30
b. Select the 7th element of the vector
c. Select all elements except the 15th. Hint: use a minus sign
d. Select the 2nd and 5th element of the vector
e. Select only the odd valued elements of `vec`.

```r
# a.
vec <- seq(11,30)
# b.
vec[7]
```

```
[1] 17
```

```r
# c.
vec[-15]
```

```
 [1] 11 12 13 14 15 16 17 18 19 20 21 22 23 24 26 27 28 29 30
```

```r
# d.
vec[c(2,5)]
```

```
[1] 12 15
```

```r
# e.
index <- seq(1,length(vec),by=2)
vec[index]
```

```
 [1] 11 13 15 17 19 21 23 25 27 29
```

Exercise 2: some calculations

a. Use the elementary functions `/`, `-`, `^` and the function `sum` to calculate the mean

$$\bar{x} = \sum_{i=1}^{n} x_i/n$$

and the standard deviation

$$\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2/(n-1)}$$

of the fare paid by the titanic passengers.

b. Verify your answer by using the built-in R functions `mean` and `sd`.

We create the data set that we used in the web based interface.

```r
titanic <- data.frame(
    pclass=rep(c("1st","2nd","3rd"),c(4,2,4)),
    survived=c(1,0,0,0,1,1,0,0,0,0),
    name=c("Allison, Master. Hudson Trevor","Dulles, Mr. William Crothers","McCarthy,
    age= c(0.9167,39,54,47,27,13,13,17,NA,NA),
    fare=c(151.55,29.7,51.8625,34.0208,13.8583,19.5,20.25,8.6625,7.75,23.4))
```

```r
# a.
fare <- titanic$fare
MeanFare <- sum(fare)/length(fare)
MeanFare
```

```
[1] 36.05541
```

```r
# standard deviation (in three baby steps)
numerator <- sum((fare-MeanFare)^2)
denominator <- (length(fare)-1)
StdFare  <- sqrt(numerator/denominator)
StdFare
```

```
[1] 42.63766
```

```r
# b.
mean(fare)
```

```
[1] 36.05541
```

```r
sd(fare)
```

```
[1] 42.63766
```

RStudio

Some useful Keyboard Shortcuts in RStudio for Windows

Over time it is useful to learn a couple of RStudio shortcuts that you frequently use. You can find an overview of them under the **Help → Keyboard Shortcuts help** menu. More shortcuts and tips can be found on the Appsilon website.

Note

On the mac, Ctrl → Cmd, Alt → Option, and Enter → Return.

- **Ctrl+Enter**: Run current line
- **Ctrl+Alt+I**: Insert new chunk
- **Alt+-**: Insert assignment `<-`
- **F1**: Show function help page when cursor is on function name
- **Ctrl+Tab**: Switch between tabs in Source Editor
- **Ctrl+1 / Ctrl+2**: Move focus to Source/Console
- **Tab / Ctrl+Space**: Code suggestion/completion
- **In Console**:
  - Arrows **Up/Down** to navigate earlier commands
  - **Ctrl+Up** pops up a listing of latest commands

Create a folder on your computer for the exercises in this R course. Open RStudio and create a new New Project in that folder. Open a new R Script file via the menu:

File → New File → R Script…

Write the code in that R Script file.

Exercise 3: importing data from Excel and first check

We will work with a dataset that gives the survival status of passengers on the Titanic. We provide the dataset in **Excel** format (Titanic3.xlsx). See this link for a description of the variables. Our data set has three additional columns: `dob` (date of birth), `family` (whether the person travelled with family) and `agecat` (age categorized in age ranges).

Download the dataset to the raw data subfolder of your project folder. Import the dataset by clicking on **Import Dataset** from the **Environment** pane. Browse for the Excel file.

The first 50 rows of the data set are shown. Look at the code under **Code Preview**. Assign the R object name `titanic` to the dataset under **Import Options** and observe that the code under **Code Preview** has changed.

Look at the mode of variable age. A problem with the mode for age is that missing values are coded as a character value `"NA"` in our Excel file; it is not recognized as an R type NA value. In the future, you may encounter other representations of missing values, such as `missing`, `404`, `-99` or `unknown`, which should also be treated as `NA`. Specify that NA represents missing values under the import options. Observe how the code under

*Code Preview **changed again. Copy the code from** Code Preview**. Click on the "Import" button. Paste the code you have copied earlier to your R Script File.

a. Inspect the data set in spreadsheet-like format that shows up after importing (if it does not show up, you can click on the `titanic` object in the **Environment** pane). Do the variables make sense? Are there any weird values? Sort the data in ascending order of age by clicking on the **age** in the data set.

b. Use the function `dim` to find the number of rows (passengers) and columns (variables) in the data set.

```
dim(titanic)
```

```
[1] 1309   17
```

Answer: 1309 rows and 17 colums

c. Use the `str` function to obtain a summary of the basic characteristics per variable. Have a look at the `survived` column. Does it have the mode that you expect? (You can also use the `mode` function to check the modes.)

```
str(titanic)
```

```
tibble [1,309 x 17] (S3: tbl_df/tbl/data.frame)
 $ pclass   : chr [1:1309] "1st" "1st" "1st" "1st" ...
 $ survived : num [1:1309] 1 1 0 0 0 1 1 0 1 0 ...
 $ name     : chr [1:1309] "Allen, Miss. Elisabeth Walton" "Allison, Master. Hudson Tr
 $ sex      : chr [1:1309] "female" "male" "female" "male" ...
 $ age      : num [1:1309] 29 0.917 2 30 25 ...
 $ sibsp    : num [1:1309] 0 1 1 1 1 0 1 0 2 0 ...
 $ parch    : num [1:1309] 0 2 2 2 2 0 0 0 0 0 ...
 $ ticket   : chr [1:1309] "24160" "113781" "113781" "113781" ...
 $ fare     : num [1:1309] 211 152 152 152 152 ...
 $ cabin    : chr [1:1309] "B5" "C22 C26" "C22 C26" "C22 C26" ...
 $ embarked : chr [1:1309] "Southampton" "Southampton" "Southampton" "Southampton" ...
 $ boat     : chr [1:1309] "2" "11" NA NA ...
 $ body     : num [1:1309] NA NA NA 135 NA NA NA NA NA 22 ...
 $ home_dest: chr [1:1309] "St Louis, MO" "Montreal, PQ / Chesterville, ON" "Montreal,
 $ dob      : num [1:1309] -28019 -17762 -18158 -28384 -26558 ...
 $ family   : chr [1:1309] "no" "yes" "yes" "yes" ...
 $ agecat   : chr [1:1309] "(18,30]" "(0,18]" "(0,18]" "(18,30]" ...
```

```r
mode(titanic$survived)
```

```
[1] "numeric"
```

`survived` is of mode numeric, while one would expect it to be a categorical variable of mode character.

Exercise 4: some columns summarized

a. Issue the commands

```r
head(titanic, 10)
```

```
# A tibble: 10 x 17
   pclass survived name     sex      age sibsp parch ticket   fare cabin embarked
   <chr>     <dbl> <chr>    <chr>  <dbl> <dbl> <dbl> <chr>   <dbl> <chr> <chr>
 1 1st           1 Allen, ~ fema~ 29         0     0 24160   211.  B5    Southam~
 2 1st           1 Allison~ male   0.917     1     2 113781 152.  C22 ~ Southam~
 3 1st           0 Allison~ fema~  2         1     2 113781 152.  C22 ~ Southam~
 4 1st           0 Allison~ male  30         1     2 113781 152.  C22 ~ Southam~
 5 1st           0 Allison~ fema~ 25         1     2 113781 152.  C22 ~ Southam~
 6 1st           1 Anderso~ male  48         0     0 19952   26.5 E12   Southam~
 7 1st           1 Andrews~ fema~ 63         1     0 13502   78.0 D7    Southam~
 8 1st           0 Andrews~ male  39         0     0 112050   0   A36   Southam~
 9 1st           1 Appleto~ fema~ 53         2     0 11769   51.5 C101  Southam~
10 1st           0 Artagav~ male  71         0     0 PC 17~  49.5 <NA>  Cherbou~
# i 6 more variables: boat <chr>, body <dbl>, home_dest <chr>, dob <dbl>,
#   family <chr>, agecat <chr>
```

```r
tail(titanic, 10)
```

```
# A tibble: 10 x 17
   pclass survived name      sex     age sibsp parch ticket  fare cabin embarked
   <chr>     <dbl> <chr>     <chr> <dbl> <dbl> <dbl> <chr>  <dbl> <chr> <chr>
 1 3rd           0 Yasbeck,~ male   27       1     0 2659   14.5  <NA>  Cherbou~
 2 3rd           1 Yasbeck,~ fema~  15       1     0 2659   14.5  <NA>  Cherbou~
 3 3rd           0 Youseff,~ male   45.5     0     0 2628    7.22 <NA>  Cherbou~
```

```
 4 3rd            0 Yousif, ~ male   NA      0     0 2647     7.22 <NA>  Cherbou~
 5 3rd            0 Youssef~ male   NA      0     0 2627    14.5  <NA>  Cherbou~
 6 3rd            0 Zabour, ~ fema~  14.5    1     0 2665    14.5  <NA>  Cherbou~
 7 3rd            0 Zabour, ~ fema~  NA      1     0 2665    14.5  <NA>  Cherbou~
 8 3rd            0 Zakarian~ male   26.5    0     0 2656     7.22 <NA>  Cherbou~
 9 3rd            0 Zakarian~ male   27      0     0 2670     7.22 <NA>  Cherbou~
10 3rd            0 Zimmerma~ male   29      0     0 315082  7.88 <NA>  Southam~
# i 6 more variables: boat <chr>, body <dbl>, home_dest <chr>, dob <dbl>,
#   family <chr>, agecat <chr>
```

What do you observe?

Answer: First and last to rows are shown. Not all columns are shown.

b. The `readxl` package that is used to import the data stores the data as a `tibble`, which is a special type of format on top of the `data.frame` format to store data in R. Override the `tibble` format using the function `as.data.frame`.

```
titanic <- as.data.frame(titanic)
```

Run the `head` and `tail` functions again. What has changed?

```
head(titanic, 10)
```

```
   pclass survived                           name    sex     age sibsp parch
1     1st        1   Allen, Miss. Elisabeth Walton female 29.0000     0     0
2     1st        1  Allison, Master. Hudson Trevor   male  0.9167     1     2
3     1st        0    Allison, Miss. Helen Loraine female  2.0000     1     2
4     1st        0 Allison, Mr. Hudson Joshua Crei   male 30.0000     1     2
5     1st        0 Allison, Mrs. Hudson J C (Bessi female 25.0000     1     2
6     1st        1             Anderson, Mr. Harry   male 48.0000     0     0
7     1st        1 Andrews, Miss. Kornelia Theodos female 63.0000     1     0
8     1st        0          Andrews, Mr. Thomas Jr   male 39.0000     0     0
9     1st        1 Appleton, Mrs. Edward Dale (Cha female 53.0000     2     0
10    1st        0         Artagaveytia, Mr. Ramon   male 71.0000     0     0
     ticket     fare   cabin    embarked boat body
1     24160 211.3375      B5 Southampton    2   NA
2    113781 151.5500 C22 C26 Southampton   11   NA
3    113781 151.5500 C22 C26 Southampton <NA>   NA
4    113781 151.5500 C22 C26 Southampton <NA>  135
```

```
5     113781 151.5500 C22 C26 Southampton <NA>   NA
6      19952  26.5500     E12 Southampton    3   NA
7      13502  77.9583      D7 Southampton   10   NA
8     112050   0.0000     A36 Southampton <NA>   NA
9      11769  51.4792    C101 Southampton    D   NA
10 PC 17609  49.5042    <NA>   Cherbourg <NA>   22
                         home_dest       dob family  agecat
1                      St Louis, MO -28019.25     no (18,30]
2  Montreal, PQ / Chesterville, ON -17761.82    yes  (0,18]
3  Montreal, PQ / Chesterville, ON -18157.50    yes  (0,18]
4  Montreal, PQ / Chesterville, ON -28384.50    yes (18,30]
5  Montreal, PQ / Chesterville, ON -26558.25    yes (18,30]
6                      New York, NY -34959.00     no (40,80]
7                        Hudson, NY -40437.75    yes (40,80]
8                       Belfast, NI -31671.75     no (30,40]
9              Bayside, Queens, NY -36785.25    yes (40,80]
10             Montevideo, Uruguay -43359.75     no (40,80]
```

```r
tail(titanic, 10)
```

```
     pclass survived                           name    sex  age sibsp parch
1300    3rd        0            Yasbeck, Mr. Antoni   male 27.0     1     0
1301    3rd        1 Yasbeck, Mrs. Antoni (Selini Al female 15.0     1     0
1302    3rd        0           Youseff, Mr. Gerious   male 45.5     0     0
1303    3rd        0              Yousif, Mr. Wazli   male   NA     0     0
1304    3rd        0          Yousseff, Mr. Gerious   male   NA     0     0
1305    3rd        0           Zabour, Miss. Hileni female 14.5     1     0
1306    3rd        0          Zabour, Miss. Thamine female   NA     1     0
1307    3rd        0      Zakarian, Mr. Mapriededer   male 26.5     0     0
1308    3rd        0            Zakarian, Mr. Ortin   male 27.0     0     0
1309    3rd        0             Zimmerman, Mr. Leo   male 29.0     0     0
     ticket    fare cabin    embarked boat body home_dest       dob family
1300   2659 14.4542  <NA>   Cherbourg    C   NA      <NA> -27288.75    yes
1301   2659 14.4542  <NA>   Cherbourg <NA>   NA      <NA> -22905.75    yes
1302   2628  7.2250  <NA>   Cherbourg <NA>  312      <NA> -34045.88     no
1303   2647  7.2250  <NA>   Cherbourg <NA>   NA      <NA>        NA     no
1304   2627 14.4583  <NA>   Cherbourg <NA>   NA      <NA>        NA     no
1305   2665 14.4542  <NA>   Cherbourg <NA>  328      <NA> -22723.12    yes
1306   2665 14.4542  <NA>   Cherbourg <NA>   NA      <NA>        NA    yes
1307   2656  7.2250  <NA>   Cherbourg <NA>  304      <NA> -27106.12     no
```

```
1308   2670  7.2250  <NA>   Cherbourg <NA>   NA      <NA> -27288.75     no
1309 315082  7.8750  <NA> Southampton <NA>   NA      <NA> -28019.25     no
      agecat
1300 (18,30]
1301  (0,18]
1302 (40,80]
1303    <NA>
1304    <NA>
1305  (0,18]
1306    <NA>
1307 (18,30]
1308 (18,30]
1309 (18,30]
```

Answer: All columns are shown.

c. Summarize the whole dataset using the `summary` function. What do you observe for each of the variables?

```r
summary(titanic)
```

```
      pclass          survived            name                 sex
 Length   :1309   Min.   :0.000   Length   :1309    Length   :1309
 N.unique :   3   1st Qu.:0.000   N.unique :1307    N.unique :   2
 N.blank  :   0   Median :0.000   N.blank  :   0    N.blank  :   0
 Min.nchar:   3   Mean   :0.382   Min.nchar:  12    Min.nchar:   4
 Max.nchar:   3   3rd Qu.:1.000   Max.nchar:  33    Max.nchar:   6
                  Max.   :1.000

     age              sibsp              parch              ticket
 Min.   : 0.1667   Min.   :0.0000   Min.   :0.000   Length   :1309
 1st Qu.:21.0000   1st Qu.:0.0000   1st Qu.:0.000   N.unique : 929
 Median :28.0000   Median :0.0000   Median :0.000   N.blank  :   0
 Mean   :29.8811   Mean   :0.4989   Mean   :0.385   Min.nchar:   3
 3rd Qu.:39.0000   3rd Qu.:1.0000   3rd Qu.:0.000   Max.nchar:  18
 Max.   :80.0000   Max.   :8.0000   Max.   :9.000
 NAs    :263
      fare              cabin             embarked              boat
 Min.   :  0.000   Length   :1309   Length   :1309    Length   :1309
 1st Qu.:  7.896   N.unique : 186   N.unique :   3    N.unique :  27
```

```
 Median : 14.454   N.blank  :    0   N.blank  :    0   N.blank  :    0  
 Mean   : 33.295   Min.nchar:    1   Min.nchar:    9   Min.nchar:    1  
 3rd Qu.: 31.275   Max.nchar:   15   Max.nchar:   11   Max.nchar:    7  
 Max.   :512.329   NAs      :1014   NAs      :    2   NAs      : 823  
 NAs    :1                                                              
      body          home_dest              dob                family         
 Min.   :  1.0   Length    :1309   Min.   :-46647   Length    :1309  
 1st Qu.: 72.0   N.unique : 368   1st Qu.:-31672   N.unique :   2  
 Median :155.0   N.blank  :    0   Median :-27654   N.blank  :   0  
 Mean   :160.8   Min.nchar:    5   Mean   :-28341   Min.nchar:   2  
 3rd Qu.:256.0   Max.nchar:   31   3rd Qu.:-25097   Max.nchar:   3  
 Max.   :328.0   NAs      : 564   Max.   :-17488                     
 NAs    :1188                      NAs    :263                       
       agecat     
 Length    :1309  
 N.unique :    4  
 N.blank  :    0  
 Min.nchar:    6  
 Max.nchar:    7  
 NAs      : 263  
```

Many are of class and mode character, and the `summary` function does not give any information. The `dob` variable has negative values.

d. Compute the 5%, 25%, 50%, 75% and 95% quantiles, the inter-quartile range and the standard deviation for age and fare. You may have to take a look at the `help` files for the functions `quantile`, `IQR` and `sd`.

```
## some quantiles
quantile(titanic$age, probs=c(0.05,0.25,0.5,0.75,0.95), na.rm=TRUE)
```

```
 5% 25% 50% 75% 95% 
  5  21  28  39  57 
```

```
quantile(titanic$fare, probs=c(0.05,0.25,0.5,0.75,0.95), na.rm=TRUE)
```

```
      5%      25%      50%      75%      95% 
  7.2250   7.8958  14.4542  31.2750 133.6500 
```

```r
## the IQR (note that this is a single number)
IQR(titanic$age, na.rm=TRUE)
```

```
[1] 18
```

```r
IQR(titanic$fare, na.rm=TRUE)
```

```
[1] 23.3792
```

```r
## the the standard deviation
sd(titanic$age, na.rm = TRUE)
```

```
[1] 14.4135
```

```r
sd(titanic$fare, na.rm = TRUE)
```

```
[1] 51.75867
```

e. For categorical variables, `summary` is not very informative. Use the `table` function to summarize the variables `pclass`, and `sex`. Also give a two-by-two table for sex and survival status using the same `table` function. Again, have a look at the help file if needed.

```r
## basic tables
table(titanic$pclass)
```

```

1st 2nd 3rd 
323 277 709 
```

```r
table(titanic$sex)
```

```

female   male 
   466    843 
```

```r
table(titanic$sex, titanic$survived)
```

```

           0   1
  female 127 339
  male   682 161
```

```r
## argument useNA has no added value because there are no missings in sex and surviv
table(titanic$sex, titanic$survived, useNA = "always")
```

```

           0   1 <NA>
  female 127 339    0
  male   682 161    0
  <NA>     0   0    0
```

f. The function **addmargins** adds the row sums and the column sums to the table of survival status by sex. The function **proportions** tabulates proportions instead of absolute numbers.

```r
addmargins(table(titanic$sex, titanic$survived))
```

```

            0    1  Sum
  female  127  339  466
  male    682  161  843
  Sum     809  500 1309
```

```r
proportions(table(titanic$sex, titanic$survived))
```

```

                  0          1
  female 0.09702063 0.25897632
  male   0.52100840 0.12299465
```

Use the `proportions` function to compute the fraction (or percentage) of survivors by sex.

```r
## basic tables
proportions(table(titanic$sex, titanic$survived), margin = 1)
```

```

                 0         1
  female 0.2725322 0.7274678
  male   0.8090154 0.1909846
```

Exercise 5: work with subsets

a. Take a look at the name, and the home town/destination of all passengers who were older than 70 years. Use the appropriate selection functions.

```
subset(titanic,age>70)[,c("name","home_dest")]
```

```
                                 name                home_dest
10            Artagaveytia, Mr. Ramon     Montevideo, Uruguay
15    Barkworth, Mr. Algernon Henry W           Hessle, Yorks
62     Cavendish, Mrs. Tyrell William Little Onn Hall, Staffs
136         Goldschmidt, Mr. George B            New York, NY
728              Connors, Mr. Patrick                    <NA>
1236              Svensson, Mr. Johan                    <NA>
```

It is seen that there is one person from Uruguay in this group. Select the single record from that person. Did this person travel with relatives?

```
subset(titanic, name=="Artagaveytia, Mr. Ramon")
```

```
   pclass survived                    name  sex age sibsp parch   ticket
10    1st        0 Artagaveytia, Mr. Ramon male  71     0     0 PC 17609
      fare cabin  embarked boat body           home_dest       dob family
10 49.5042  <NA> Cherbourg <NA>   22 Montevideo, Uruguay -43359.75     no
    agecat
10 (40,80]
```

Answer: the person did not travel with relatives

b. Make a table of survivor status by sex, but only for the first class passengers. What do you observe? Compare this with the survival status for the 3rd class passengers.

```
first <- xtabs(~sex+survived, data=titanic, subset=(pclass=="1st"))
first
```

```
        survived
sex        0   1
  female   5 139
  male   118  61
```

```
third <- xtabs(~sex+survived, data=titanic, subset=(pclass=="3rd"))
third
```

```
        survived
sex        0   1
  female 110 106
  male   418  75
```

Answer: females were much more likely to survive than males, especially among those who travelled first class

Exercise 6: make factor variables

a. Make passenger class and sex into factor variables.
b. Add a variable `status` to the Titanic data set, which gives the survival status of the passengers as a factor variable with labels "no" and "yes" describing whether individuals survived. Make the value "no" the first level.

```
titanic$status <- factor(titanic$survived, labels=c("no","yes"))
```

c. Run the `summary` function again. What change do you observe compared to Exercise 4?

```
summary(titanic)
```

```
      pclass            survived             name                sex           
 Length    :1309   Min.    :0.000   Length    :1309   Length    :1309  
 N.unique  :   3   1st Qu.:0.000   N.unique  :1307   N.unique  :   2  
 N.blank   :   0   Median :0.000   N.blank   :   0   N.blank   :   0  
 Min.nchar:   3   Mean    :0.382   Min.nchar:   12   Min.nchar:   4  
 Max.nchar:   3   3rd Qu.:1.000   Max.nchar:   33   Max.nchar:   6  
                  Max.    :1.000                                      

      age                sibsp               parch               ticket      
 Min.    : 0.1667   Min.    :0.0000   Min.    :0.000   Length    :1309  
 1st Qu.:21.0000   1st Qu.:0.0000   1st Qu.:0.000   N.unique  : 929  
 Median :28.0000   Median :0.0000   Median :0.000   N.blank   :   0  
```

```
 Mean   :29.8811   Mean   :0.4989   Mean   :0.385   Min.nchar:   3
 3rd Qu.:39.0000   3rd Qu.:1.0000   3rd Qu.:0.000   Max.nchar:  18
 Max.   :80.0000   Max.   :8.0000   Max.   :9.000
 NAs    :263
      fare              cabin           embarked           boat
 Min.   :  0.000   Length    :1309   Length    :1309   Length    :1309
 1st Qu.:  7.896   N.unique : 186   N.unique :   3   N.unique :  27
 Median : 14.454   N.blank  :   0   N.blank  :   0   N.blank  :   0
 Mean   : 33.295   Min.nchar:   1   Min.nchar:   9   Min.nchar:   1
 3rd Qu.: 31.275   Max.nchar:  15   Max.nchar:  11   Max.nchar:   7
 Max.   :512.329   NAs      :1014   NAs      :   2   NAs      : 823
 NAs    :1
      body          home_dest           dob              family
 Min.   :  1.0   Length    :1309   Min.   :-46647   Length    :1309
 1st Qu.: 72.0   N.unique : 368   1st Qu.:-31672   N.unique :   2
 Median :155.0   N.blank  :   0   Median :-27654   N.blank  :   0
 Mean   :160.8   Min.nchar:   5   Mean   :-28341   Min.nchar:   2
 3rd Qu.:256.0   Max.nchar:  31   3rd Qu.:-25097   Max.nchar:   3
 Max.   :328.0   NAs      : 564   Max.   :-17488
 NAs    :1188                     NAs    :263
       agecat      status
 Length    :1309   no :809
 N.unique :   4   yes:500
 N.blank  :   0
 Min.nchar:   6
 Max.nchar:   7
 NAs      : 263
```

Answer: Passenger class and sex are summarized properly, and status is summarized as a categorical variable.

Exercise 7: working with dates

a. The variable `dob` gives the date of birth, using days since January 1st, 1960 (this is the time origin used by Stata). Transform this to a true date value using the function `as.Date` (specify the origin argument).

```r
titanic$dob <- as.Date(titanic$dob, origin = "1960/1/1")
```

b. The default display is "year/month/day". Make a table of the day of the month that the passengers were born, using the `format` function. What do you observe?

```r
table(format(titanic$dob, "%d"))
```

```
 13  14  15  16 
 15 940  86   5 
```

Answer: All have a day in the middle of the month

c. What was the earliest date of birth among all the passengers on the boat. Does this correspond to the age of the oldest person on the date that the ship sank (April 15th, 1912)? Use the functions `min` and `max`.

```r
min(titanic$dob, na.rm = TRUE)
```

```
[1] "1832-04-14"
```

```r
max(titanic$age, na.rm = TRUE)
```

```
[1] 80
```

```r
min(titanic$dob, na.rm = TRUE) + max(titanic$age, na.rm = TRUE) * 365.25
```

```
[1] "1912-04-15"
```

Yes, it does correspond